A Balance Law Interpretation of the Fokker-Planck Equation

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Overview

Heat a small spot on a metal bar and the hot spot spreads and smooths. Drop a speck of dye into still water and it does the same. Now let a particle be carried along by a known differential equation while being jostled at random: our knowledge of where it is — a probability density — spreads and smooths in just the same way. Why do all three spread by the same law? Because all three are governed by the same piece of bookkeeping, a *balance law*: whatever accumulates in a region must have crossed its boundary.

In this paper we play this one game on ever larger boards. First we use a balance law to derive the heat equation, in one and then in several dimensions. Next we use the same balance law — with probability in place of heat, and with regions that move with the flow of a differential equation — to derive the equation governing the propagation of a probability density function driven by that differential equation. Finally, we observe that a balance law is linear in the flux, so fluxes add: crediting a moving region with both mechanisms of transfer at once — probability carried by the flow, and probability diffusing through the boundary as heat does — yields the Fokker-Planck equation.

Derivation of the Heat Equation

We examine the problem of heat flow in R^n. First we look at the problem in one dimension and then see how it generalizes to arbitrary dimensions. The strategy is to write a balance law for the heat in an arbitrary region in R^n based on the conservation of energy. This will lead to a local expression of heat flow; namely, a partial differential equation. By relating heat to temperature, an equation for the temperature is obtained. The resulting partial differential equation for temperature is called *the heat equation*.

The derivation is a four step template: balance the quantity of interest in an arbitrary region over an arbitrary time interval; shrink the time interval to get an instantaneous balance; use the arbitrariness of the region to localize — converting an integral statement into a pointwise one; and close with an assumption relating the flux to the quantity being balanced. Every equation in this paper — the Fokker-Planck equation included — comes from replaying these four steps on a bigger board.

The Heat Equation in One Dimension

We derive a partial differential equation for the temperature distribution in a one dimensional material. The reader has likely already solved the lumped version of this problem: Newton's law of cooling, in which the whole object has a single temperature $T(t)$ satisfying $\dot{T} = -r(T - T_a)$. The heat equation is what replaces Newton's law when the temperature is allowed to vary from point to point, so that the object must do its bookkeeping locally. To do this we derive a balance law for the heat contained in an arbitrary interval $[x_1, x_2]$

over an arbitrary time interval $[t_1, t_2]$. Let $h(t, x)$ be the heat per unit length at time t and position x. Let $f(t, x)$ be the heat flux at time t and position x, where heat flux is defined as the rate at which heat flows †. The increase (or decrease) in the heat of the interval $[x_1, x_2]$ over $[t_1, t_2]$ is equal to the integral of the rate at which heat has moved into (or out of) the region through the boundaries. Mathematically, this balance may be expressed as:

$$\int_{x_1}^{x_2} h(t_2, x) - h(t_1, x)\; dx = \int_{t_1}^{t_2} f(t, x_1) - f(t, x_2)\; dt$$

Now, divide this expression by $(t_2 - t_1)$ and take the limit as $t_2 \to t_1$. This gives:

$$\int_{x_1}^{x_2} h_t(t_1, x)\; dx = f(t_1, x_1) - f(t_1, x_2) \tag{1}$$

This can be written as

$$\frac{d}{dt}\int_{x_1}^{x_2} h(t, x)\; dx\bigg|_{t=t_1} = f(t_1, x_1) - f(t_1, x_2) \tag{2}$$

Equation (2) can be interpreted as an instantaneous balance law for the heat contained in the interval $[x_1, x_2]$. It states that the rate of change of the heat in the interval $[x_1, x_2]$ is equal to the rate at which heat enters through the boundary.

By the *Fundamental Theorem of Calculus*, $\int_{x_1}^{x_2} f_x(t_1, x)\, dx = f(t_1, x_2) - f(t_1, x_1)$. Therefore, bringing the right hand side to the left hand side in equation (1) we may write:

$$\int_{x_1}^{x_2} h_t(t_1, x) + f_x(t_1, x)\; dx = 0$$

If we *assume* that the integrand is continuous, then since the interval $[x_1, x_2]$ is arbitrary, the integrand must vanish for all x; that is,

$$h_t(t_1, x) + f_x(t_1, x) = 0$$

Finally, since t_1 is arbitrary the above must be true for all times t; so that,

$$h_t(t, x) + f_x(t, x) = 0$$

To get an equation for the temperature, u, we must relate it to the heat and the heat flux. To do so, we must make *assumptions* about properties of the material. The simplest assumption about the heat is that it is proportional to temperature:† $h(t, x) = c\,u(t, x)$. This proportionality constant, $c > 0$, is called the specific heat. The simplest assumption about the heat flux is that it is proportional to the temperature gradient: $f(t, x) = -ku_x(t, x)$ with $k > 0$. Therefore, we may write:

$$c\,u_t - ku_{xx} = 0$$

Letting $\kappa = k/c$ we obtain the one dimensional heat equation with diffusion constant κ:

$$\frac{\partial u}{\partial t} = \kappa \frac{\partial^2 u}{\partial x^2}.$$

† Positive heat flux f means heat flowing in the direction of increasing x.

† The temperature of a substance is the average kinetic energy of the material's molecules. When we refer to heat we mean the heat density, the amount of heat energy per unit volume. The constant c used here is the *volumetric* heat capacity (energy per unit volume per kelvin), equal to $\rho\, c_p$ where ρ is the mass density and c_p is the specific heat per unit mass. We will follow common practice and refer to c as the specific heat, with this caveat understood.

The larger the value of κ, the more rapid the diffusion. Since κ is made from two constants, we give an interpretation to understand their role. Consider a bar made of differing materials. With differing materials we have two independent mechanisms controlling heat flow: the heat capacity of any portion of the bar controls the value of the specific heat, c; and, the insulating property of the material controls the heat flux rate, k.

Under these conditions, we can get a better physical understanding of the influence of c and k on κ. If the bar behaves more like water at a certain spot, it can absorb more heat and therefore is less diffusive there. If the bar is made of a material that is more insulating at a certain spot, like styrofoam, the heat flux rate constant goes down and therefore the material is less diffusive there. (Varying cross-sectional thickness manifests in the 1-D model only through its effect on c and k.)

We note that c and k may depend on x. In this case we obtain a generalized heat equation:

$$\frac{\partial u}{\partial t} = \frac{1}{c(x)}\frac{\partial}{\partial x}\left(k(x)\frac{\partial u}{\partial x}\right)$$

Technically, to have similar properties of the heat equation, $c(x)$ and $k(x)$ should satisfy (for all x):

1. $c(x) > c_0$ for some $c_0 > 0$.
2. $k(x) > k_0$ for some $k_0 > 0$.
3. $c(x)$ is continuous and $k(x)$ is continuously differentiable with respect to x.

The Heat Equation in Higher Dimensions

We derive a partial differential equation for the temperature distribution of a material in R^n. To do this we will mimic each of the steps in the last section. We start by deriving a balance law for the heat contained in an arbitrary region Ω over an arbitrary time interval $[t_1, t_2]$. Let $h(t, \mathbf{x})$ be the heat per unit volume at time t and position $\mathbf{x}$. Let $\mathbf{f}(t, \mathbf{x})$ be the heat flux at time t and position $\mathbf{x}$, where heat flux is defined as a vector describing the rate at which heat flows per area†. The increase (or decrease) in the heat of the region Ω over $[t_1, t_2]$ is equal to the integral of the rate at which heat has moved into (or out of) the region through the boundaries. Mathematically, this may be expressed as:

$$\int_\Omega h(t_2, \mathbf{x}) - h(t_1, \mathbf{x})\, d\mathbf{x} = -\int_{t_1}^{t_2} \underbrace{\oint_{\partial\Omega} \overbrace{\mathbf{f}(t, \mathbf{x}) \cdot \mathbf{n}}^{\text{Heat flow rate per unit area leaving } \Omega}\, d\mathbf{S}}_{\text{Heat flow rate leaving } \Omega}\, dt$$

As in the one dimensional case, divide by $(t_2 - t_1)$ and take the limit as $t_2 \to t_1$. This gives:

$$\int_\Omega h_t(t_1, \mathbf{x})\, d\mathbf{x} = -\oint_{\partial\Omega} \mathbf{f}(t_1, \mathbf{x}) \cdot \mathbf{n}\, d\mathbf{S} \tag{3}$$

This can be written as

$$\frac{d}{dt}\int_\Omega h(t, \mathbf{x})\, d\mathbf{x}\bigg|_{t=t_1} = -\oint_{\partial\Omega} \mathbf{f}(t_1, \mathbf{x}) \cdot \mathbf{n}\, d\mathbf{S} \tag{4}$$

Equation (4) can be interpreted as an instantaneous balance law for the heat contained in the region Ω. It states that the rate of change of the heat in the region Ω is equal to the rate at which heat enters through the boundary.

† By "area" we mean the $(n-1)$-dimensional measure of the boundary $\partial\Omega$.

By the *Divergence Theorem*, $\int_\Omega \nabla \cdot \mathbf{f}(t_1, \mathbf{x})\, d\mathbf{x} = \oint_{\partial\Omega} \mathbf{f}(t_1, \mathbf{x}) \cdot \mathbf{n}\, d\mathbf{S}$. Therefore, bringing the right hand side to the left hand side in equation (3) we may write:

$$\int_\Omega h_t(t_1, \mathbf{x}) + \nabla \cdot \mathbf{f}(t_1, \mathbf{x})\; d\mathbf{x} = 0$$

If we *assume* that the integrand is continuous, then since the region Ω is arbitrary, the integrand must vanish for all $\mathbf{x}$; that is,

$$h_t(t_1, \mathbf{x}) + \nabla \cdot \mathbf{f}(t_1, \mathbf{x}) = 0$$

Finally, since t_1 is arbitrary the above must be true for all times t; so that,

$$h_t(t, \mathbf{x}) + \nabla \cdot \mathbf{f}(t, \mathbf{x}) = 0$$

To get an equation for the temperature, u, we must relate it to the heat and the heat flux. To do so, we must make *assumptions* about the properties of the material. The simplest assumption about the heat is that it is proportional to temperature: $h(t, \mathbf{x}) = c\, u(t, \mathbf{x})$. As in the one dimensional case, the proportionality constant, $c > 0$, is called the specific heat. The simplest assumption about the heat flux is that it is proportional to the temperature gradient:[‡] $\mathbf{f}(t, \mathbf{x}) = -K \nabla u(t, \mathbf{x})$ with $K > 0$. Therefore, we may write:

$$c\, u_t(t, \mathbf{x}) - K \nabla \cdot \nabla u(t, \mathbf{x}) = 0$$

Letting $\kappa = K/c$ and noting that $\Delta = \nabla \cdot \nabla$, we obtain the n-dimensional heat equation with diffusion constant κ:

$$\frac{\partial u}{\partial t} = \kappa\, \Delta u.$$

We note that c and K may depend on $\mathbf{x}$ and we note K could be a matrix. In this case we obtain a generalized heat equation:

$$\frac{\partial u}{\partial t} = \frac{1}{c(\mathbf{x})} \nabla \cdot \Big(K(\mathbf{x}) \nabla u \Big)$$

Technically, to have similar properties of the heat equation, $c(\mathbf{x})$ and $K(\mathbf{x})$ should satisfy (for all $\mathbf{x}$):

1. $c(\mathbf{x}) > c_0$ for some $c_0 > 0$;
2. $K(\mathbf{x})$ should be a positive definite symmetric matrix;
3. The smallest eigenvalue of $K(\mathbf{x})$ is greater than k_0 for some $k_0 > 0$;
4. $c(\mathbf{x})$ is continuous and $K(\mathbf{x})$ is continuously differentiable with respect to $\mathbf{x}$.

When $K(\mathbf{x})$ is a matrix, the heat flux vector $\mathbf{f}(t, \mathbf{x})$ will not, in general, point in the direction opposite of the maximum temperature increase. This may be interpreted physically as having a material that has a different heat flow rate in different directions. In this case, the heat flux vector, representing the heat flow path, will be modified by these material differences and will not necessarily flow along the temperature gradient.

Propagation of Probability Density Functions

In this section, we wish to derive equations for the propagation of probability density functions driven by systems of differential equations.

‡ That is, the *direction* of heat flow should be in the opposite direction of maximum change of temperature; and, the *magnitude* of the heat flow should be proportional to the magnitude of the maximum change in the temperature.

A First Order System

Given a first order system of differential equations of the form:

$$\dot{\mathbf{x}} = \mathbf{f}(t, \mathbf{x}) \tag{5a}$$

$$\mathbf{x}(0) = \mathbf{x}_0 \tag{5b}$$

let $\mathcal{X}(t; \mathbf{x}_0)$ be the solution; which is assumed to be invertible on the interval $[0, T)$. That is, for each $t \in [0, T)$, $\mathcal{X}(t; \cdot) : R^n \to R^n$ is invertible. If X is a random variable with probability density function $P_X(\mathbf{x})$, then we wish to compute the probability density function of the random variable induced by applying the transformation $\mathcal{X}(t; \cdot)$ to the random variable X. This will be done by writing down a balance law for a suitable probabilistic expression. As in previous sections, we will not obtain a formula for the induced probability density function; rather, we will produce a partial differential equation describing its evolution.

A Balance Law Computation

Given an arbitrary region, $\widehat{\Omega}$, and an arbitrary time $t \in [0, T)$, the probability contained in $\widehat{\Omega}$ will remain the same as the region is transformed either forward or backward in time under the differential equation (5a-b). Therefore, there is a one parameter family of regions Ω_τ such that $\Omega_t = \widehat{\Omega}$ and $\Omega_0 = \Omega$. That is, Ω morphs into $\widehat{\Omega}$ under the action of the differential equation (5a-b). Let $p(t, \mathbf{x}) = P_{\mathcal{X}(t;X)}(\mathbf{x})$ be the probability density function of the induced random variable $\mathcal{X}(t; X)$. The conservation of probability of $P_{\mathcal{X}(t;X)}(\Omega_t)$ is expressed mathematically as an instantaneous balance law:

$$\frac{d}{d\tau} \int_{\Omega_\tau} p(\tau, \mathbf{x})\, d\mathbf{x} = 0 \tag{6}$$

The new difficulty, compared with the heat sections, is that the region itself moves. We handle it by changing the representation: change the integration variable so as to keep the region constant, differentiate, and change back. To this end, let $\mathbf{x} = \mathcal{X}(\tau, \mathbf{z})$. The integral becomes:

$$\frac{d}{d\tau} \int_{\Omega} p(\tau, \mathcal{X}(\tau, \mathbf{z})) \left| \frac{\partial \mathcal{X}(\tau,\mathbf{z})}{\partial \mathbf{z}} \right| d\mathbf{z} = 0$$

which is

$$\int_{\Omega} \big(p_\tau(\tau, \mathcal{X}(\tau, \mathbf{z})) + \nabla p(\tau, \mathcal{X}(\tau, \mathbf{z})) \cdot \dot{\mathcal{X}}(\tau, \mathbf{z})\big) \left| \frac{\partial \mathcal{X}(\tau,\mathbf{z})}{\partial \mathbf{z}} \right| + p(\tau, \mathcal{X}(\tau, \mathbf{z})) \frac{d}{d\tau} \left| \frac{\partial \mathcal{X}(\tau,\mathbf{z})}{\partial \mathbf{z}} \right| d\mathbf{z} = 0 \tag{7}$$

It turns out that the derivative, $\frac{d}{d\tau} \left| \frac{\partial \mathcal{X}(\tau,\mathbf{z})}{\partial \mathbf{z}} \right|$, may be written as†

$$\frac{d}{d\tau} \left| \frac{\partial \mathcal{X}(\tau,\mathbf{z})}{\partial \mathbf{z}} \right| = \left| \frac{\partial \mathcal{X}(\tau,\mathbf{z})}{\partial \mathbf{z}} \right| \operatorname{tr}\big(\nabla \mathbf{f}(\tau, \mathcal{X}(\tau, \mathbf{z}))\big) = \left| \frac{\partial \mathcal{X}(\tau,\mathbf{z})}{\partial \mathbf{z}} \right| \nabla \cdot \mathbf{f}(\tau, \mathcal{X}(\tau, \mathbf{z}))$$

Therefore, (7) becomes

$$\int_{\Omega} \Big(p_\tau(\tau, \mathcal{X}(\tau, \mathbf{z})) + \nabla p(\tau, \mathcal{X}(\tau, \mathbf{z})) \cdot \mathbf{f}(\tau, \mathcal{X}(\tau, \mathbf{z})) + p(\tau, \mathcal{X}(\tau, \mathbf{z})) \nabla \cdot \mathbf{f}(\tau, \mathcal{X}(\tau, \mathbf{z}))\Big) \left| \frac{\partial \mathcal{X}(\tau,\mathbf{z})}{\partial \mathbf{z}} \right| d\mathbf{z} = 0 \tag{8}$$

† This is Liouville's formula, and it is four lines from tools we already have. First, differentiating (5a) with respect to $\mathbf{z}$ gives $\frac{d}{d\tau} \frac{\partial \mathcal{X}}{\partial \mathbf{z}} = \nabla \mathbf{f}(\tau, \mathcal{X}(\tau, \mathbf{z})) \frac{\partial \mathcal{X}}{\partial \mathbf{z}}$. Second, Jacobi's formula for differentiating a determinant states $\frac{d}{d\tau} \det M = \det M \operatorname{tr}\big(M^{-1} \frac{dM}{d\tau}\big)$. Applying Jacobi's formula with $M = \frac{\partial \mathcal{X}}{\partial \mathbf{z}}$ and substituting the first fact, the factors M and M^{-1} cancel inside the trace, leaving $\operatorname{tr}\big(\nabla \mathbf{f}\big)$.

Changing coordinates back, (8) becomes

$$\int_{\Omega_\tau} p_\tau(\tau,\mathbf{x}) + \nabla p(\tau,\mathbf{x})\cdot\mathbf{f}(\tau,\mathbf{x}) + p(\tau,\mathbf{x})\nabla\cdot\mathbf{f}(\tau,\mathbf{x})\ d\mathbf{x} = 0$$

This may be written as

$$\int_{\Omega_\tau} p_\tau(\tau,\mathbf{x}) + \nabla\cdot\big(p(\tau,\mathbf{x})\mathbf{f}(\tau,\mathbf{x})\big)\ d\mathbf{x} = 0 \tag{9}$$

Equation (9) is true for all $\tau \in [0,T)$, in particular it is true for $\tau = t$. Assuming the integrand in (9) is continuous, and since $\widehat{\Omega} = \Omega_t$ and t are both arbitrary, we have for all $t \in [0,T)$ and for all $\mathbf{x}$

$$p_t(t,\mathbf{x}) + \nabla\cdot\big(p(t,\mathbf{x})\mathbf{f}(t,\mathbf{x})\big) = 0 \tag{10a}$$

$$p(0,\mathbf{x}) = P_X(\mathbf{x}) \tag{10b}$$

In coordinates this is written as

$$\frac{\partial p(t,\mathbf{x})}{\partial t} + \sum_{i=1}^{n}\frac{\partial}{\partial x_i}\big(p(t,\mathbf{x})f_i(t,\mathbf{x})\big) = 0$$

$$p(0,\mathbf{x}) = P_X(\mathbf{x})$$

Equations (10a-b) constitute a first-order initial-value problem for the probability density function $P_X(\mathbf{x})$ with respect to the differential equation (5a-b).

A Second Order System

Many physical systems are more naturally described by a system of second order differential equations than a first order system. In this section, we derive an equation governing the evolution of a probability density function driven by a second order system. Consider the second order system of differential equations of the form:

$$\ddot{\mathbf{x}} = \mathbf{f}(t,\mathbf{x},\dot{\mathbf{x}}) \tag{11a}$$

$$\mathbf{x}(0) = \mathbf{x}_0 \tag{11b}$$

$$\dot{\mathbf{x}}(0) = \mathbf{v}_0 \tag{11c}$$

This may be written as a first order system as follows: Let $\mathbf{v} = \dot{\mathbf{x}}$, then (11a-c) may be written as

$$\dot{\mathbf{x}} = \mathbf{v} \tag{12a}$$

$$\dot{\mathbf{v}} = \mathbf{f}(t,\mathbf{x},\mathbf{v}) \tag{12b}$$

$$\mathbf{x}(0) = \mathbf{x}_0 \tag{12c}$$

$$\mathbf{v}(0) = \mathbf{v}_0 \tag{12d}$$

We can use the result of the last section to write a formula for the propagation of the probability density function. Let $P_{X,V}(\mathbf{x},\mathbf{v})$ be the joint probability density function for random variables X and V. Let $p(t,\mathbf{x},\mathbf{v}) = P_{\mathcal{X}(t;X,V),\mathcal{V}(t;X,V)}(\mathbf{x},\mathbf{v})$ denote the probability density function of the pair (X,V), transformed by the solution of (12a-d) at time t (here we evaluate the deterministic flow at the random initial conditions, producing random variables $\mathcal{X}(t;X,V)$ and $\mathcal{V}(t;X,V)$ with joint density p). From the previous section, p satisfies

$$p_t(t,\mathbf{x},\mathbf{v}) + (\,\nabla_{\mathbf{x}}p(t,\mathbf{x},\mathbf{v})\ \ ,\ \nabla_{\mathbf{v}}p(t,\mathbf{x},\mathbf{v})\,)\cdot\begin{pmatrix}\mathbf{v}\\ \mathbf{f}(t,\mathbf{x},\mathbf{v})\end{pmatrix} + p(t,\mathbf{x},\mathbf{v})\ \mathrm{tr}\begin{pmatrix}0 & I\\ \nabla_{\mathbf{x}}\mathbf{f}(t,\mathbf{x},\mathbf{v}) & \nabla_{\mathbf{v}}\mathbf{f}(t,\mathbf{x},\mathbf{v})\end{pmatrix} = 0$$

Simplifying and adding the initial condition this becomes

$$p_t(t,\mathbf{x},\mathbf{v}) + \nabla_{\mathbf{x}} p(t,\mathbf{x},\mathbf{v})\cdot\mathbf{v} + \nabla_{\mathbf{v}} p(t,\mathbf{x},\mathbf{v})\cdot\mathbf{f}(t,\mathbf{x},\mathbf{v}) + p(t,\mathbf{x},\mathbf{v})\nabla_{\mathbf{v}}\cdot\mathbf{f}(t,\mathbf{x},\mathbf{v}) = 0 \tag{13a}$$

$$p(0,\mathbf{x},\mathbf{v}) = P_{X,V}(\mathbf{x},\mathbf{v}) \tag{13b}$$

Or, using the product rule $\nabla\cdot(\varphi\mathbf{w}) = \nabla\varphi\cdot\mathbf{w} + \varphi\nabla\cdot\mathbf{w}$ to combine the last two terms,

$$p_t(t,\mathbf{x},\mathbf{v}) + \nabla_{\mathbf{x}} p(t,\mathbf{x},\mathbf{v})\cdot\mathbf{v} + \nabla_{\mathbf{v}}\cdot\big(p(t,\mathbf{x},\mathbf{v})\mathbf{f}(t,\mathbf{x},\mathbf{v})\big) = 0 \tag{14a}$$

$$p(0,\mathbf{x},\mathbf{v}) = P_{X,V}(\mathbf{x},\mathbf{v}) \tag{14b}$$

In coordinates this is written as

$$\frac{\partial p(t,\mathbf{x},\mathbf{v})}{\partial t} + \sum_{i=1}^{n}\frac{\partial p(t,\mathbf{x},\mathbf{v})}{\partial x_i}v_i + \sum_{i=1}^{n}\frac{\partial}{\partial v_i}\big(p(t,\mathbf{x},\mathbf{v})f_i(t,\mathbf{x},\mathbf{v})\big) = 0 \tag{15a}$$

$$p(0,\mathbf{x},\mathbf{v}) = P_{X,V}(\mathbf{x},\mathbf{v}) \tag{15b}$$

The Fokker-Planck Equation I

We now consider the evolution of a probability density function driven primarily by the differential system (5a-b), but which incorporates the dynamics of diffusion as well. Such a governing equation is called a Fokker-Planck equation. Note that $p(t,\mathbf{x})$ can no longer be the transported density of the last section: for that density, equation (6) says the probability in a moving region never changes, which is exactly what diffusion violates. Instead, let X be a random variable with density $P_X(\mathbf{x})$, let $\mathcal{X}(t;\mathbf{x}_0)$ be the solution of (5a-b), and let $p(t,\mathbf{x})$ denote the (as yet unknown) probability density of the state at time t, whose primary dynamics is the flow of (5a-b). We replace assumption (6): a moving region now also gains or loses probability by diffusion through its boundary. As before, let $\widehat{\Omega}$ be an arbitrary region in R^n and let Ω_τ parameterize a family of regions such that $\Omega_0 = \Omega$ and $\Omega_t = \widehat{\Omega}$.

Instead of equation (6), we assume that

$$\frac{d}{d\tau}\int_{\Omega_\tau} p(\tau,\mathbf{x})\,d\mathbf{x} = \oint_{\partial\Omega_\tau}\big(A(\mathbf{x})\nabla p(\tau,\mathbf{x})\big)\cdot\mathbf{n}\,d\mathbf{S} \tag{16}$$

In the simplest case, $A(\mathbf{x}) = I$, equation (16) says that the changing regions, Ω_τ, allow diffusion based on the probability gradient on the boundary. This is the same mechanism as the heat equation. In the general case, the diffusion behaves as the diffusion in the generalized heat equation.

The computation (7)–(9) of the last section nowhere used the fact that p was the transported density: it is an identity, valid for any smooth density p, expressing the derivative of the left hand side of (16) when the regions Ω_τ move with the flow of (5a-b). Reusing it, we may write

$$\int_{\Omega_\tau} p_\tau(\tau,\mathbf{x}) + \nabla\cdot\big(p(\tau,\mathbf{x})\mathbf{f}(\tau,\mathbf{x})\big)\,d\mathbf{x} = \oint_{\partial\Omega_\tau}\big(A(\mathbf{x})\nabla p(\tau,\mathbf{x})\big)\cdot\mathbf{n}\,d\mathbf{S} \tag{17}$$

Using the *Divergence Theorem* on the right hand side of (17) and combining with the left hand side, we have

$$\int_{\Omega_\tau} p_\tau(\tau,\mathbf{x}) + \nabla\cdot\big(p(\tau,\mathbf{x})\mathbf{f}(\tau,\mathbf{x})\big) - \nabla\cdot\big(A(\mathbf{x})\nabla p(\tau,\mathbf{x})\big)\,d\mathbf{x} = 0 \tag{18}$$

Since this is true for all $\tau \in [0,t]$; and, t and $\Omega_t = \widehat{\Omega}$ are arbitrary, we may write the Fokker-Planck equation governing the evolution of $P_X(\mathbf{x})$†.

$$p_t(t,\mathbf{x}) + \nabla \cdot \big(p(t,\mathbf{x})\mathbf{f}(t,\mathbf{x})\big) = \nabla \cdot \big(A(\mathbf{x})\nabla p(t,\mathbf{x})\big) \tag{19a}$$

$$p(0,\mathbf{x}) = P_X(\mathbf{x}) \tag{19b}$$

In coordinates this is written as

$$\frac{\partial p(t,\mathbf{x})}{\partial t} + \sum_{i=1}^{n} \frac{\partial}{\partial x_i}\Big(p(t,\mathbf{x})f_i(t,\mathbf{x})\Big) = \sum_{i=1}^{n} \frac{\partial}{\partial x_i}\Big(\sum_{j=1}^{n} A_{ij}(\mathbf{x})\frac{\partial p(t,\mathbf{x})}{\partial x_j}\Big) \tag{20a}$$

$$p(0,\mathbf{x}) = P_X(\mathbf{x}) \tag{20b}$$

A remark on the form of the diffusion term is in order. The balance law (16) models the diffusive flux as $-A(\mathbf{x})\nabla p$ — Fick's law, exactly as in the heat equation — which produces the diffusion term $\nabla \cdot \big(A(\mathbf{x})\nabla p\big)$. The Fokker-Planck equation associated with a stochastic differential equation is often written instead with diffusion term $\sum_{i,j} \frac{\partial^2}{\partial x_i \partial x_j}\big(D_{ij}(\mathbf{x})\,p\big)$. The two forms agree when the diffusion matrix is constant; when it varies with $\mathbf{x}$ they differ by a term of first order, which may be absorbed into the drift $\mathbf{f}$. The form derived here is the one dictated by a Fickian flux assumption.

The Fokker-Planck Equation II

We now consider the evolution of a probability density function driven primarily by the second order differential system (12a-d), but which incorporates the dynamics of diffusion as well. If X and V are random variables with joint density $P_{X,V}(\mathbf{x},\mathbf{v})$, and $\{\mathcal{X}(t;\mathbf{x}_0,\mathbf{v}_0)\ ,\ \mathcal{V}(t;\mathbf{x}_0,\mathbf{v}_0)\}$ is the pair of functions representing the solution of (12a-d), then — as in the last section — let $p(t,\mathbf{x},\mathbf{v})$ denote the (as yet unknown) joint probability density of the state $(\mathbf{x},\mathbf{v})$ at time t, whose primary dynamics is the flow of (12a-d). As before, let $\widehat{\Omega}$ be an arbitrary region in $R^n \times R^n$, and let Ω_τ parameterize a family of regions in $R^n \times R^n$ such that $\Omega_0 = \Omega$ and $\Omega_t = \widehat{\Omega}$.

We assume, as in the previous section, that the primary dynamics of the evolution of $p(t,\mathbf{x},\mathbf{v})$ is the differential system (12a-d). Additionally, we assume that probability diffuses through the $\mathbf{v}$ component of the regions Ω_τ based *only* on the gradient of p with respect to $\mathbf{v}$. That is, the instantaneous balance law (6) becomes

$$\frac{d}{d\tau}\int_{\Omega_\tau} p(\tau,\mathbf{x},\mathbf{v})\,d\mathbf{x}\,d\mathbf{v} = \oint_{\partial\Omega_\tau} \big(\,0\ \ ,\ A(\mathbf{x},\mathbf{v})\nabla_{\mathbf{v}} p(\tau,\mathbf{x},\mathbf{v})\,\big)\cdot\begin{pmatrix}\mathbf{n}_x\\ \mathbf{n}_v\end{pmatrix}\,d\mathbf{S} \tag{21}$$

Here $(\mathbf{n}_x,\mathbf{n}_v)$ is the outward normal to the boundary of Ω_τ, a surface in $R^n \times R^n$ with surface measure $d\mathbf{S}$. The left hand side has been previously calculated; so, using the *Divergence Theorem* on the right hand side, we may write

$$\begin{aligned}\int_{\Omega_\tau} p_\tau(\tau,\mathbf{x},\mathbf{v}) + \nabla_{\mathbf{x}} p(\tau,\mathbf{x},\mathbf{v})\cdot\mathbf{v} + \nabla_{\mathbf{v}}\cdot\big(p(\tau,\mathbf{x},\mathbf{v})\mathbf{f}(\tau,\mathbf{x},\mathbf{v})\big)&\\ -\nabla_{\mathbf{v}}\cdot\big(A(\mathbf{x},\mathbf{v})\nabla_{\mathbf{v}} p(\tau,\mathbf{x},\mathbf{v})\big)\,d\mathbf{x}\,d\mathbf{v} = 0&\end{aligned} \tag{22}$$

† We make the same assumptions about $A(\mathbf{x})$ (in this section and the next) that were made when deriving the heat equation in n dimensions:

1. $A(\mathbf{x})$ should be a positive definite symmetric matrix;

2. The smallest eigenvalue of $A(\mathbf{x})$ is greater than a_0 for some $a_0 > 0$;

3. $A(\mathbf{x})$ is continuously differentiable with respect to $\mathbf{x}$.

Again, since $\widehat{\Omega} = \Omega_t$ and t are arbitrary, the Fokker-Planck equation governing the evolution of $P_{X,V}(\mathbf{x}, \mathbf{v})$ is

$$p_t(t, \mathbf{x}, \mathbf{v}) + \nabla_{\mathbf{x}} p(t, \mathbf{x}, \mathbf{v}) \cdot \mathbf{v} + \nabla_{\mathbf{v}} \cdot \big(p(t, \mathbf{x}, \mathbf{v})\mathbf{f}(t, \mathbf{x}, \mathbf{v})\big) = \nabla_{\mathbf{v}} \cdot \big(A(\mathbf{x}, \mathbf{v})\nabla_{\mathbf{v}} p(t, \mathbf{x}, \mathbf{v})\big) \tag{23a}$$

$$p(0, \mathbf{x}, \mathbf{v}) = P_{X,V}(\mathbf{x}, \mathbf{v}) \tag{23b}$$

The coordinates for the right-hand side were written in the last section, so we may write the coordinate version of this Fokker-Planck equation as

$$\frac{\partial p(t, \mathbf{x}, \mathbf{v})}{\partial t} + \sum_{i=1}^{n} \frac{\partial p(t, \mathbf{x}, \mathbf{v})}{\partial x_i} v_i + \sum_{i=1}^{n} \frac{\partial}{\partial v_i}\big(p(t, \mathbf{x}, \mathbf{v}) f_i(t, \mathbf{x}, \mathbf{v})\big) = \sum_{i=1}^{n} \frac{\partial}{\partial v_i}\left(\sum_{j=1}^{n} A_{ij}(\mathbf{x}, \mathbf{v}) \frac{\partial p(t, \mathbf{x}, \mathbf{v})}{\partial v_j}\right) \tag{24a}$$

$$p(0, \mathbf{x}, \mathbf{v}) = P_{X,V}(\mathbf{x}, \mathbf{v}) \tag{24b}$$

In the case when $A(\mathbf{x}, \mathbf{v}) = \kappa\mathcal{A}$, $\mathcal{A}$ a constant positive definite symmetric matrix, (24a-b) becomes

$$\frac{\partial p(t, \mathbf{x}, \mathbf{v})}{\partial t} + \sum_{i=1}^{n} \frac{\partial p(t, \mathbf{x}, \mathbf{v})}{\partial x_i} v_i + \sum_{i=1}^{n} \frac{\partial}{\partial v_i}\big(p(t, \mathbf{x}, \mathbf{v}) f_i(t, \mathbf{x}, \mathbf{v})\big) = \kappa \sum_{i=1}^{n}\sum_{j=1}^{n} \mathcal{A}_{ij} \frac{\partial^2 p(t, \mathbf{x}, \mathbf{v})}{\partial v_i \, \partial v_j} \tag{25a}$$

$$p(0, \mathbf{x}, \mathbf{v}) = P_{X,V}(\mathbf{x}, \mathbf{v}) \tag{25b}$$

If we further specialize $\mathcal{A}$ to the identity matrix, then (25a-b) becomes

$$\frac{\partial p(t, \mathbf{x}, \mathbf{v})}{\partial t} + \sum_{i=1}^{n} \frac{\partial p(t, \mathbf{x}, \mathbf{v})}{\partial x_i} v_i + \sum_{i=1}^{n} \frac{\partial}{\partial v_i}\big(p(t, \mathbf{x}, \mathbf{v}) f_i(t, \mathbf{x}, \mathbf{v})\big) = \kappa \sum_{i=1}^{n} \frac{\partial^2 p(t, \mathbf{x}, \mathbf{v})}{\partial v_i^2} \tag{26a}$$

$$p(0, \mathbf{x}, \mathbf{v}) = P_{X,V}(\mathbf{x}, \mathbf{v}) \tag{26b}$$

In the one dimensional case, this is further simplified to

$$\frac{\partial p(t, x, v)}{\partial t} + \frac{\partial p(t, x, v)}{\partial x} v + \frac{\partial}{\partial v}\big(p(t, x, v) f(t, x, v)\big) = \kappa \frac{\partial^2 p(t, x, v)}{\partial v^2} \tag{27a}$$

$$p(0, x, v) = P_{X,V}(x, v) \tag{27b}$$

One Game, Five Boards

Every equation in this paper came from the same four steps: balance a quantity in an arbitrary region; shrink the time interval to get an instantaneous balance; use the arbitrariness of the region to localize; and close with an assumption about the flux. Only the board changed.

Board	**Flux through the boundary**	**Resulting equation**
interval $[x_1, x_2]$	$-ku_x$	$u_t = \kappa u_{xx}$
region $\Omega \subset R^n$	$-K\nabla u$	$u_t = \kappa\, \Delta u$
moving region Ω_τ	none: the region rides the flow	$p_t + \nabla \cdot (p\mathbf{f}) = 0$
moving region Ω_τ	$-A\nabla p$	$p_t + \nabla \cdot (p\mathbf{f}) = \nabla \cdot (A\nabla p)$
phase space $(\mathbf{x}, \mathbf{v})$	$-A\nabla_{\mathbf{v}} p$, in $\mathbf{v}$ only	equation (23a)

This table answers the question posed in the Overview. The hot spot on the bar, the speck of dye, and the randomly jostled particle spread by the same law because they are governed by the same bookkeeping: whatever accumulates in a region must have crossed its boundary. And the Fokker-Planck equation, which can appear as a pronouncement when first met, is from this point of view not a new kind of equation at all. A balance law is linear in the flux, so fluxes add: credit a moving region with the flow's transport and with a Fickian diffusion through its boundary, and the Fokker-Planck equation is what the four steps produce. The reader who can replay the four steps of Section 2 owns every equation in this paper.